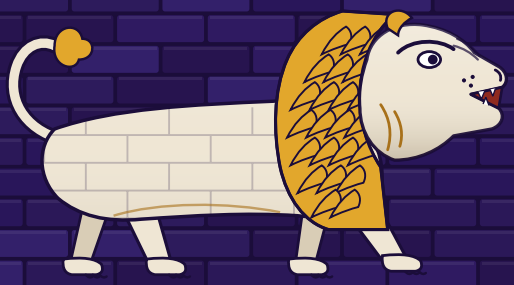
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

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OPTICS AND WAVES

A handout of ideas, facts and problems



12 sections · Edition of November 2026 · 41 pages

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How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Optics and Waves

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

1 Wave language and simple harmonic motion

Aims. Convert an oscillation count into frequency and period; Relate displacement, speed and acceleration at key SHM positions.
Before this section: One oscillation is one complete cycle; reciprocal quantities; displacement from equilibrium.

A wave carries a disturbance and energy. Mechanical waves need a medium; electromagnetic waves can travel through vacuum. Transverse displacement is perpendicular to travel; longitudinal displacement is parallel. Simple harmonic motion has a restoring acceleration proportional and opposite to displacement.

Idea 1. One complete oscillation returns the motion to the same position and direction. Period asks how long that return takes; frequency asks how many returns occur each second. They describe the same repeating motion from opposite viewpoints.

Fact 1.

$$T = 1/f$$
$$f = N/t$$
$$a = -\omega^2 x$$

where N is the number of complete cycles (dimensionless); t is the total elapsed time (s); f is the cycles per second (Hz); T is the time for one cycle (s); ω is the angular frequency (rad/s).

Why it holds. More cycles within the same time means larger frequency. A larger frequency leaves less time for each cycle, so period decreases.

Method 1.

1. Count complete cycles rather than individual crossings of equilibrium.
2. Divide cycle count by total time for frequency.
3. Find period as total time per cycle or the reciprocal of frequency.
4. For SHM compare the point with equilibrium and use restoring acceleration toward equilibrium.

Recognising it. A counted number of complete oscillations over a measured time signals a frequency and period calculation.

Pr 1. If 20 oscillations take 5 s, frequency is 4 Hz and period is 0.25 s.

Solution. Twenty complete oscillations in 5 s means $20/5 = 4$ cycles per second, so $f = 4$ Hz. One cycle takes $5/20 = 0.25$ s, giving $T = 0.25$ s. At an extreme the object must turn around, so its instantaneous speed is zero. Its displacement magnitude is largest there; restoring acceleration therefore has maximum magnitude and points toward equilibrium.

Common error. The direction of particle motion is not necessarily the direction of wave travel.

In the examination. Write $f = N/t$ and $T = t/N$ side by side to prevent interchanging hertz and seconds.

Pr 2. At SHM equilibrium is acceleration magnitude maximum?

Pr 3. A source makes 50 complete oscillations in 10 s. Find frequency and period.

Pr 4. A source doubles its period. How does its frequency change?

(a) It halves

(b) It doubles

(c) It stays unchanged

2 Read a travelling wave without deriving it

Aims. Read amplitude, wavelength and frequency from a travelling-wave expression; Use $v = f\lambda$ and distinguish pattern motion from particle motion.

Before this section: Sine-wave shape; π; coefficients of x and t; SI units.

Amplitude is maximum displacement, wavelength is crest-to-crest separation, and frequency counts cycles per second. Wave speed is the speed of the pattern, not the oscillating particle. In a sine wave, the coefficients of position and time reveal wavelength and period.

Idea 2. The source decides how often a wave repeats, while the medium sets its pattern speed under the stated conditions. Wavelength is the distance the pattern advances during one cycle. At fixed speed, more cycles each second mean each cycle occupies less distance.

Fact 2.

$$v = f\lambda$$
$$k = 2\pi/\lambda$$
$$\omega = 2\pi f$$
$$y = A\sin(kx - \omega t + \phi)$$

where A is the amplitude or maximum displacement (m); k is the wave number (rad/m); ω is the angular frequency (rad/s); λ is the wavelength (m); f is the frequency (Hz).
Conditions. The example takes x and y in metres and t in seconds.

Why it holds. Frequency times wavelength gives distance travelled each second. In the same nondispersive medium, a frequency increase changes wavelength while speed stays fixed.

Method 2.

1. Match the wave to y = A sin(kx - ωt + ϕ).
2. Read A directly, then divide 2π by k to obtain wavelength.
3. Divide ω by 2π to obtain frequency.
4. Multiply frequency by wavelength for pattern speed and check the units.

Recognising it. Use the fixed-speed relationship when frequency changes but the same nondispersive medium is retained.

Pr 5. For $y = 0.02\sin(4\pi x - 20\pi t)$, amplitude is 0.02 m, wavelength is 0.50 m, frequency is 10 Hz and speed is 5 m/s.

Solution. Compare y = 0.02 sin(4πx - 20πt) with the standard form: A = 0.02 m, k = 4π rad/m and $\omega = 20\pi$ rad/s. Wavelength is $2\pi/(4\pi) = 0.50$ m. Frequency is $20\pi/(2\pi) = 10$ Hz. Pattern speed is $10 \times 0.50 = 5$ m/s; this is not the maximum transverse speed of an individual particle.

Common error. The coefficient of time is angular frequency, not ordinary frequency.

In the examination. Identify coefficients before calculating; treating k as wavelength or ω as ordinary frequency loses the factor 2π.

Pr 6. Is the coefficient 20π in the example the frequency in hertz?

Pr 7. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

Pr 8. A wave enters a medium where its speed is lower while frequency stays fixed. Its wavelength:

(a) Must remain fixed

(b) Becomes shorter

(c) Becomes longer

Fact 3. Phase means position within a cycle. A phase difference tells whether two oscillations are at the same or different stages of a cycle.

$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda} \quad \text{(phase-difference magnitude at one time)}$$

$$\Delta\phi = 2\pi\frac{\Delta t}{T} \quad \text{(phase-change magnitude at one place)}$$

1. One full cycle is 2π radians. Half a cycle is π radians.
2. At a fixed time, points one wavelength apart are in phase. Points half a wavelength apart are opposite in phase.
3. At one place, a full period brings the oscillation back to the same phase.
4. These formulas give the size of the phase difference. Its sign depends on the order of comparison and the travelling-wave convention.

Pr 9. Two points on a sinusoidal wave are separated by $\lambda/4$. Find their phase-difference magnitude at the same time.

3 Wavefronts, spreading and group velocity

Aims. Predict intensity and amplitude changes for spherical spreading; Distinguish wavefront, ray, phase velocity and group velocity.
Before this section: Surface area grows with radius squared; square roots; amplitude versus intensity.

A ray indicates propagation direction and is normal to a wavefront. Plane wavefronts are flat, spherical wavefronts spread from a point source, and cylindrical wavefronts spread around a line source. Phase velocity tracks a crest; group velocity tracks the envelope of a narrow wave packet.

Idea 3. Picture fixed source power being shared over an expanding spherical shell. The larger shell carries the same total power when nothing is absorbed, but each square metre receives a smaller share. Geometric spreading can therefore reduce intensity without dissipating the total power.

Fact 4.

$$I \propto A^2$$

$$I_{\text{spherical}} \propto 1/r^2$$

where I is the energy-transfer rate per unit area ($\mathrm{W/m^2}$); A is the wave amplitude (depends on disturbance); r is the distance from point source (m).
Conditions. The spreading example assumes an isotropic point source, no absorption and unchanged medium properties.

Why it holds. Sphere area grows with radius squared. With total power fixed, intensity decreases by the same area factor.

Method 3.

1. Identify whether the source produces plane, spherical or cylindrical spreading.
2. For an isotropic point source use intensity proportional to $1/r^2$.
3. Convert an intensity ratio to an amplitude ratio by taking its square root.
4. In description questions, identify a crest for phase velocity and the packet envelope for group velocity.

Recognising it. An isotropic point source with negligible absorption permits inverse-square spreading; other geometries need their own model.

Pr 10. For an isotropic point source without absorption, doubling distance reduces intensity to one quarter and amplitude to one half.

Solution. The same source power crosses a sphere around the source at each distance. Doubling radius makes the sphere area four times as large. The available power per square metre is therefore one quarter as large. Because intensity is proportional to amplitude squared, the amplitude ratio is $\sqrt{(1/4)} = 1/2$.

Common error. The ideal plane-wave model does not have spherical inverse-square spreading.

In the examination. Do not apply the inverse-square intensity rule automatically to a plane wave or line source.

Pr 11. An intensity of 18 W/m^2 is measured at r. What is it at 3r?

Pr 12. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

Pr 13. Without absorption, which quantity stays the same on complete spheres farther from an ideal point source?

(a) Intensity per square metre

(b) Wave amplitude at each point

(c) Total power crossing each sphere

4 Superposition, standing waves and beats

Aims. Calculate beat frequency from two nearby frequencies; Connect fixed-string harmonics with allowed wavelengths.
Before this section: Adding displacements; wavelength; frequency; fixed endpoints.

Overlapping disturbances add algebraically. In-phase equal waves reinforce; opposite-phase equal waves cancel. A standing wave forms when matching waves travel in opposite directions. Nodes stay at zero displacement. Beats are slow changes in loudness from nearby frequencies.

Idea 4. A string fixed at both ends must have nodes at its endpoints. Only patterns fitting whole numbers of loops between them can persist as normal modes. One loop spans half a wavelength, so a string length is not automatically a complete wavelength.

Fact 5.

$$f_{\text{beat}} = |f_1 - f_2|$$
$$\lambda_n = 2L/n \text{ (string fixed at both ends)}$$

where f_{beat} is the number of loudness cycles per second (Hz); L is the vibrating string length (m); n is the harmonic number, 1, 2, 3, ... (dimensionless); λ_n is the wavelength of harmonic n (m).

Why it holds. Fixed endpoints restrict which wavelengths fit. At fixed wave speed, fitting more loops shortens wavelength and increases frequency.

Method 4.

1. For superposition add instantaneous signed displacements at the same point.
2. For beats take the positive difference of the two nearby frequencies.
3. For a string fixed at both ends draw nodes at both ends.
4. Fit n half-wavelengths into the length, then use wave speed to find each frequency.

Recognising it. For a string fixed at both ends, draw its node-and-loop pattern before selecting the harmonic wavelength.

Pr 14. Two tones at 256 and 260 Hz produce 4 beats per second.

Solution. The tones have frequencies 256 Hz and 260 Hz. Their frequency difference is 4 Hz, so there are four loudness beats each second. For the separate string example, the fundamental has one loop between two end nodes. That loop occupies half a wavelength, so $\lambda 1 = 2\text{L}$ and $\text{f1} = \text{v}/(2\text{L})$.

Common error. A node is not a point where two travelling waves are individually absent.

In the examination. A beat frequency is a difference, not the average of the two tone frequencies.

Pr 15. For a fixed string, how does the second-harmonic frequency compare with the fundamental?

Pr 16. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

Pr 17. Adjacent nodes on an ideal standing wave are separated by:

(a) Half a wavelength

(b) One wavelength

(c) One quarter wavelength

5 Light, photons and the electromagnetic picture

Aims. Compare photon energy using frequency or vacuum wavelength; Distinguish beam brightness from energy per photon.

Before this section: Direct and inverse proportionality; wave frequency; energy transfer.

Light behaves as an electromagnetic wave in propagation and exchanges energy in photons. Photon energy increases with frequency. In the photoelectric effect, emission requires enough energy per photon; increasing brightness alone cannot overcome a frequency below threshold.

Idea 5. Photon energy belongs to an individual packet, whereas intensity describes energy arriving per area each second. Increasing frequency makes each packet more energetic. Making a fixed-colour beam brighter can instead increase the number of packets arriving, without changing the energy in each.

Fact 6.

$$E_\gamma = hf = hc/\lambda_0$$
$$\lambda_0 = \text{vacuum wavelength}$$
$$c = 3.00 \times 10^8 \text{ m/s}$$

where E_γ is the energy of one photon (J); h is the Planck constant (J s); f is the light frequency (Hz); λ is the vacuum wavelength (m).

Why it holds. Photon energy increases directly with frequency. Vacuum wavelength decreases as frequency increases because vacuum propagation speed is fixed.

Method 5.

1. Decide whether the question asks about one photon or the whole beam.
2. Use Eγ proportional to frequency, or inversely proportional to vacuum wavelength.
3. For brightness comparisons separate photon energy from the number arriving each second.

4. For photoelectric emission compare energy per photon with the threshold requirement.

Recognising it. Use frequency ratios to compare photon energies and vacuum wavelengths without inserting unnecessary constants.

Pr 18. Blue light has shorter wavelength than red light and therefore higher photon energy.

Solution. Blue light has a shorter vacuum wavelength than red light. Since $E\gamma = hc/\lambda$, the smaller wavelength corresponds to a larger energy per photon. Increasing red-beam brightness at the same colour increases the energy arriving per second, usually through more photons. Each red photon still has the same lower energy; brightness alone does not change its frequency.

Common error. Higher intensity does not mean higher energy per photon. In E = hc/lambda, use the vacuum wavelength. At an ordinary stationary boundary, frequency and photon energy stay the same even though wavelength in the medium changes.

In the examination. Use the phrase energy per photon when discussing threshold effects; total beam energy does not answer the same question.

Pr 19. If light A has twice light B frequency, how do photon energy and vacuum wavelength compare?

Pr 20. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

Pr 21. Light crosses a stationary ordinary air-glass boundary. The transmitted photon energy:

(a) Increases solely because the ray bends

(b) Stays the same because frequency stays the same

(c) Decreases because wavelength in glass is shorter

6 Reflection, refraction and total internal reflection

Aims. Track speed, frequency and wavelength across a refracting boundary; Recognize the two conditions for total internal reflection.

Before this section: Normal to a surface; sine; reciprocal refractive index.

Angles of incidence, reflection and refraction are measured from the normal. At a stationary interface frequency stays the same, while speed and wavelength may change. When light enters a higher refractive index at a nonzero angle, the ray bends toward the normal. Total internal reflection needs travel from higher to lower index and an incidence angle greater than the critical angle.

Idea 6. A ray entering a lower-index medium bends away from the normal. As incidence angle increases, the transmitted direction approaches the boundary. At the critical angle the limiting transmitted ray runs along that boundary; beyond it the ordinary transmitted ray cannot propagate away into the second medium.

Fact 7.

$$\theta_i = \theta_r$$
$$n = c/v$$
$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$
$$\sin\theta_c = n_2/n_1$$

where n is the refractive index (dimensionless); c is the vacuum light speed (m/s); v is the light speed in medium (m/s); θ_c is the critical incidence angle (degrees or radians).
Conditions. The boundary is stationary and the media are treated as uniform; air index is approximated as 1.

Why it holds. Only higher-to-lower index travel can reach the critical-angle condition. Total internal reflection requires incidence above, rather than merely equal to, the critical angle.

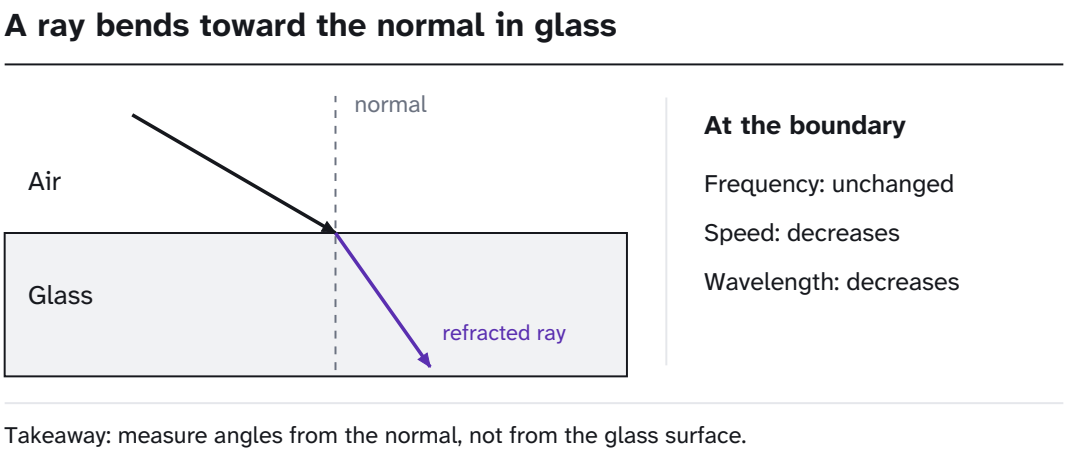


Figure 1. Figure 03. At oblique incidence from air into glass, the ray bends toward the normal. At a stationary boundary its frequency is unchanged, while speed and wavelength decrease.

Method 6.

1. Draw the surface normal and measure all ray angles from it.
2. Use v = c/n to compare speeds; frequency stays fixed at a stationary boundary.
3. Use wavelength = speed/frequency to compare wavelengths.
4. For total internal reflection check higher-to-lower index travel, then compare incidence with the critical angle.

Recognising it. The words critical angle signal a check of travel direction followed by the lower-to-higher index sine ratio.

Pr 22. From air into glass with index 1.5, light slows to 2×10^8 m/s.

Solution. Air is approximated by n = 1 and glass has n = 1.5. In glass, speed is $3.00 \times 10^8 / 1.5 = 2.00 \times 10^8$ m/s. The frequency is set by the source and remains the same across the boundary. Because speed drops while frequency stays fixed, wavelength in glass is two thirds of its air value.

Common error. The critical angle is an incidence angle in the higher-index medium.

In the examination. At the critical angle the refracted ray runs along the boundary; total internal reflection occurs above that angle.

Pr 23. Can light travelling from air into glass undergo total internal reflection at that interface?

Pr 24. Find the critical angle for glass of index 1.5 into air of index 1.0.

Pr 25. At exactly the critical angle, the refracted ray travels:

(a) Along the normal

(b) Back along the incident ray

(c) Along the boundary

7 Mirrors, lenses and image signs

Aims. Use the stated image-distance sign convention consistently; Infer image reality, orientation and relative size from di and m.

Before this section: Reciprocals; positive and negative numbers; a simple ray sketch.

A real image is formed where rays actually meet and can be projected on a screen. A virtual image is where rays appear to originate. Use one sign convention consistently: real object distance positive, real image distance positive; converging focal length positive and diverging focal length negative.

Idea 7. A lens equation locates where rays meet or appear to originate. Magnification then compares image height with object height, including orientation. A negative magnification records an inverted image; it does not mean the physical image has a negative length.

Fact 8.

$$1/f = 1/d_o + 1/d_i$$
$$m = -d_i/d_o$$

where f is the signed focal length (cm or m consistently); d_o is the object distance (same length unit as f); d_i is the signed image distance (same length unit as f); m is the transverse magnification (dimensionless).
Conditions. Use the lesson convention: real object and real image distances positive; converging focal length positive.

Why it holds. A converging lens with an object at twice focal length gives a real image at twice focal length on the other side. Magnitude of magnification gives relative size, while its sign gives orientation.

Method 7.

1. Write the sign convention and assign f and do before substitution.
2. Rearrange the lens equation to isolate 1/di and then invert.
3. Calculate m = -di/do.
4. Use di sign for real or virtual, m sign for orientation, and magnitude of m for size.

Recognising it. Use one consistent sign convention when focal length and object distance are given, then interpret the numerical result.

Pr 26. For a converging lens with focal length 10 cm and object distance 30 cm, image distance is 15 cm and magnification is -0.5: real, inverted, reduced.

Solution. A converging lens has f = +10 cm and the real object has do = +30 cm. Then 1/di = $1/10 - 1/30 = 1/15$ per cm, so di = +15 cm. Magnification is $-15/30 = -0.5$. Positive image distance means real; negative magnification means inverted; magnitude 0.5 means half the object height.

Common error. Do not call every image behind a lens virtual; interpret actual ray convergence and your convention.

In the examination. After subtraction, invert 1/di to get di; include all three requested image descriptions.

Pr 27. What does m = +2 tell you about an image?

Pr 28. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

Pr 29. Why can a real image be projected onto a screen?

(a) Actual rays meet at its position

(b) Its magnification must be positive

(c) All images lie on the lens surface

8 Optical systems, prisms and instruments

Aims. Explain the different roles of objective and eyepiece; Calculate telescope angular magnification for normal adjustment.
Before this section: Focal length; image formation; ratios of lengths.

A prism refracts light at two surfaces; dispersion separates colours because refractive index depends on wavelength. In a multi-lens system, the first image becomes the object for the next lens. A microscope and a telescope both use an objective and eyepiece, but they serve different object distances.

Idea 8. A telescope objective forms an intermediate image of a distant scene, and an eyepiece enlarges the angle under which it is viewed. Angular magnification compares viewing angles, so it is different from the height ratio for one ordinary lens image.

Fact 9.

$$m_{\text{total}} = m_1 m_2$$
$$|M_{\text{telescope}}| = f_o/f_e \ \text{(normal adjustment)}$$

where f_o is the objective focal length (cm or m consistently); f_e is the eyepiece focal length (same length unit as fo); M is the angular magnification (dimensionless); m_1, m_2 is the transverse magnifications of successive stages (dimensionless).
Conditions. The telescope is in normal adjustment, giving the final image at infinity.

Why it holds. For normal adjustment, a shorter eyepiece focal length gives larger angular magnification with the same objective. A focal-length ratio is dimensionless when both lengths use the same unit.

Method 8.

1. Identify the instrument and whether it views a distant or nearby object.
2. For a normal-adjustment astronomical telescope divide objective focal length by eyepiece focal length for magnification magnitude.
3. State image orientation separately.
4. For a general multi-lens system locate the first image before using it as the next object.

Recognising it. Use the focal-length ratio for an astronomical telescope in normal adjustment, not for an arbitrary two-lens system.

Pr 30. An astronomical telescope with objective focal length 80 cm and eyepiece 4 cm has angular magnification magnitude 20.

Solution. The telescope objective focal length is 80 cm and the eyepiece focal length is 4 cm. Their ratio is $80/4 = 20$, so the angular magnification magnitude is 20. An astronomical telescope with this usual two-converging-lens arrangement gives an inverted image. The ratio needs no length conversion because both focal lengths use the same unit.

Common error. Linear image magnification and angular magnification are different quantities.

In the examination. Use the telescope focal-length ratio only for the stated telescope model, not as a universal magnification formula.

Pr 31. Keeping the objective fixed, does doubling the eyepiece focal length increase or decrease telescope magnification?

Pr 32. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

Pr 33. Which eyepiece gives greater magnification with the same objective in normal adjustment?

(a) Both give equal magnification

(b) The shorter-focal-length eyepiece

(c) The longer-focal-length eyepiece

9 Interference and diffraction

Aims. Predict how slit geometry changes an interference or diffraction pattern; Distinguish double-slit bright fringes from single-slit dark fringes.
Before this section: Wavelength; coherent sources; nanometres and millimetres; small angles.

Interference occurs when overlapping waves combine. A stable visible pattern needs a fixed phase relationship (coherence). Diffraction is spreading associated with an opening or obstacle. Narrower slits spread light more. Young double-slit bright fringes are evenly spaced when the angles are small.

Idea 9. Coherent slits create bright and dark directions through path differences between overlapping waves. A screen turns those angles into distances. Moving the screen farther away enlarges the visible spacing even though the slit geometry and bright angular directions remain unchanged.

Fact 10.

$$\Delta y = \lambda L/d$$
$$d \sin\theta = m\lambda \text{ (double-slit bright)}$$
$$a \sin\theta = m\lambda \text{ (single-slit dark, } m \neq 0\text{)}$$

where Δy is the double-slit fringe spacing (m); L is the slit-to-screen distance (m); d is the separation of double slits (m); a is the single-slit width (m); λ is the wavelength (m).
Conditions. Even double-slit spacing uses the small-angle, distant-screen approximation and coherent illumination.

Why it holds. Longer wavelength increases the angular change needed between neighbouring bright fringes. A more distant screen maps the same angular separation to a larger

physical distance.

Method 9.

1. Identify whether the setup is double-slit interference or single-slit diffraction.
2. Label d as slit separation and a as slit width; they describe different distances.
3. For double-slit spacing use $\lambda L/d$ with all lengths in consistent units.
4. For a single slit remember that a smaller width moves the first minima outward and broadens the centre.

Recognising it. For two coherent narrow slits and a distant screen, use wavelength, slit separation and screen distance to find spacing.

Pr 34. Doubling screen distance doubles double-slit fringe spacing.

Solution. Double-slit spacing is proportional to screen distance when wavelength and slit separation stay fixed. Replacing L by 2L therefore doubles fringe spacing. In the separate single-slit comparison, the first dark direction obeys $\sin\theta = \lambda/a$. Halving a makes that angle larger, so the bright central maximum broadens.

Common error. The double-slit bright condition is not the single-slit bright condition.

In the examination. Label whether you are locating a bright double-slit fringe or a dark single-slit fringe before using the order number.

Pr 35. If double-slit separation doubles with screen distance and wavelength fixed, what happens to fringe spacing?

Pr 36. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

Pr 37. Only the double-slit screen distance doubles. The bright-fringe angular directions:

(a) Double along with screen spacing

(b) All become zero

(c) Stay unchanged while screen spacing grows

Fact 11. Recognize gratings, Michelson paths and thin films. All three appear in the physical-optics section. Start by identifying which arrangement the question describes.

$$\Delta\text{path} = 2\Delta x \quad \text{(one Michelson mirror moves)}$$

1. A diffraction grating has many equally spaced slits. It makes sharp bright directions that separate wavelengths.
2. A Michelson interferometer splits light into two arms. Each beam travels to a mirror and back, so mirror movement changes a round-trip path.
3. A thin film produces interference between reflections from its two surfaces. Reflection can add a phase reversal, so do not copy the ordinary double-slit bright rule.
4. For fringe counting in a simple Michelson setup, each one-wavelength change in path difference shifts the pattern by one fringe.

Pr 38. In a Michelson interferometer using 500 nm light in air, one mirror moves 250 nm along its arm. What path change and fringe shift result?

10 Polarization and Malus law

Aims. Apply the initial half-intensity rule to unpolarized light; Apply Malus law using the angle between polarization and analyzer axes.
Before this section: Cosine of common angles; intensity versus field amplitude; sequential fractions.

Polarization describes the electric-field direction of a transverse light wave. An ideal polarizer transmits half the intensity of unpolarized light. An analyzer then transmits a fraction set by the angle between its axis and the incident polarization. Linear, circular and elliptical polarization describe different field-tip motions.

Idea 10. The first ideal polarizer turns unpolarized light into linearly polarized light while transmitting half its intensity. The analyzer receives that output. It then passes a fraction determined by the angle to the already established polarization, so the first and second stages use different rules.

Fact 12.

$$I = I_0 \cos^2\theta$$
$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

where I_0 is the intensity incident on the analyzer ($\mathrm{W/m^2}$ or stated units); I is the intensity transmitted by analyzer (same as I0); θ is the angle between incident polarization and analyzer axis (degrees or radians).
Conditions. The calculation assumes ideal polarizers and initially unpolarized illumination.

Why it holds. The half-intensity rule applies to initially unpolarized input. For linearly polarized input, the analyzer transmits a squared-cosine fraction.

Method 10.

1. Check whether the first incident beam is unpolarized or already linearly polarized.
2. For unpolarized light through an ideal first polarizer divide intensity by two.
3. Use that transmitted intensity as I0 for the analyzer.
4. Multiply by $\cos^2\theta$, where θ is between the two relevant axes.

Recognising it. For successive polarizers, identify input polarization and process each element in order.

Pr 39. Unpolarized intensity 100 units passes a polarizer then an analyzer at 60 degrees.

Solution. The initial 100 intensity units are unpolarized, so the first ideal polarizer transmits 50 units. The analyzer axis is at 60° to the first polarizer axis. Cos 60° is 0.5 and its square is 0.25. The analyzer therefore transmits $50 \times 0.25 = 12.5$ units, one eighth of the original intensity.

Common error. Do not halve already polarized intensity again before applying Malus law.

In the examination. Square the cosine, not the angle; and do not apply an extra half factor to an already polarized analyzer input.

Pr 40. Two ideal polarizers have perpendicular axes. What does the second transmit from the first?

Pr 41. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

Pr 42. A second ideal polarizer is aligned with the first. What fraction of the first output passes it?

(a) All of it

(b) Half of it

(c) None of it

11 Scattering, apparent depth and lens power

Aims. Calculate apparent depth near normal viewing and signed lens power; Explain why small-particle scattering favours shorter wavelengths.
Before this section: Refraction; inverse relationships; centimetres-to-metres conversion.

Small particles compared with wavelength produce Rayleigh scattering, which favours shorter wavelengths strongly. Droplets comparable to or larger than visible wavelengths can scatter a broad range of colours, helping explain white clouds. Refraction makes a submerged object appear shallower when viewed near the normal from air.

Idea 11. Lens power describes how strongly a lens converges or diverges rays. A shorter focal-length magnitude means stronger bending and greater power magnitude. The negative sign of a diverging lens communicates the direction of its optical effect, rather than the physical size of the lens.

Fact 13.

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$
$$d_{\text{apparent}} \approx d_{\text{real}}/n$$
$$P_{\text{lens}} = 1/f \ \text{(f in metres)}$$

where d_{real} is the actual depth (m); d_{apparent} is the apparent depth (m); n is the liquid refractive index relative to air approximation (dimensionless); P is the lens optical power (D = m^{-1}); f is the signed focal length (m).
Conditions. The apparent-depth approximation assumes viewing from air near the normal; the Rayleigh rule applies to particles much smaller than wavelength.

Why it holds. Reciprocals make smaller focal-length magnitudes correspond to larger power magnitudes. Dioptres use metres in the focal-length reciprocal, so centimetres must be converted first.

Method 11.

1. Select the relevant model: small-particle scattering, near-normal depth or lens power.
2. For depth viewed from air divide real depth by the liquid index.
3. For power convert focal length to metres before taking its reciprocal.
4. Retain the focal-length sign so converging power is positive and diverging power negative.

Recognising it. A supplied focal length and a request for optical power call for a signed reciprocal in metre units.

Pr 43. A pool 2 m deep in water of index 1.33 appears about 1.50 m deep from air near the normal.

Solution. For the pool, near-normal apparent depth is $2/1.33 \approx 1.50$ m, so the bottom ap-

pears shallower. The index does not change the real 2 m depth; it changes the directions of the emerging rays. For the separate converging lens, $f = +0.25$ m. Its power is $1/0.25 = +4\ \mathrm{m}^{-1}$, written +4 dioptres.

Common error. A focal length of 25 cm is not 25 m; using it directly gives the wrong lens power.

In the examination. A reciprocal of a focal length in centimetres does not directly give dioptres; convert to metres first.

Pr 44. A diverging lens has focal length -50 cm. What is its power?

Pr 45. A diverging lens has focal length -50 cm. Find its optical power.

Pr 46. Which diverging lens has greater power magnitude: focal length -0.25 m or -0.50 m?

(a) They match because both diverge

(b) The -0.25 m lens

(c) The -0.50 m lens

12 Recognize advanced optics without long derivations

Aims. Recognize what a quarter-wave plate changes about polarized light; Use Brewster angle and plane-wave relations at recognition level.
Before this section: Linear and circular polarization; phase as position within an oscillation cycle; refractive index.

The wave equation describes a disturbance propagating in space and time. Complex notation is a bookkeeping method for amplitude and phase; the measured field is real. Maxwell laws connect electric and magnetic fields. Birefringent materials have different refractive behaviour for different polarizations. At Brewster incidence the reflected component can be linearly polarized.

Idea 12. Brewster incidence is a special transparent-interface geometry. In ordinary transparent nonmagnetic media, the reflected light at that angle lacks the component polarized in the incidence plane. This concerns polarization of the reflected component, not complete reflection of all incoming energy.

Fact 14.

$$\tan\theta_B = n_2/n_1$$
$$E = cB \text{ (plane wave in vacuum)}$$

where θ_B is the Brewster incidence angle from the normal (degrees or radians); n_1, n_2 is the incident and transmitted medium indices (dimensionless); E is the plane-wave electric-field magnitude (V/m); B is the plane-wave magnetic-field magnitude (T).
Conditions. The wave plate is operating at its intended wavelength; Brewster law here uses ordinary transparent nonmagnetic media.

Why it holds. The refractive-index ratio sets the tangent of the Brewster angle from the normal. Brewster incidence and total internal reflection describe different conditions and effects.

Method 12.

1. Identify whether the question concerns phase delay, interface polarization or vacuum wave fields.
2. For a quarter-wave plate, compare the two orthogonal component amplitudes and their phase difference.
3. For Brewster incidence use the index ratio with inverse tangent.
4. For a plane vacuum wave connect field magnitudes using E = cB without a long derivation.

Recognising it. Use Brewster angle when an interface question links refractive indices and linearly polarized reflection.

Pr 47. A quarter-wave plate can turn suitable linearly polarized light into circular polarization by introducing a quarter-cycle phase difference between equal orthogonal components.

Solution. Resolve suitable incident linear polarization along the plate principal axes. Equal component amplitudes are required for the stated circular-polarization result. A quarter-wave plate adds a quarter-cycle, or $\pi/2$, relative phase delay. Equal amplitudes with this delay give circular polarization; unequal amplitudes generally give elliptical polarization.

Common error. These recognition notes do not replace a lecturer-assigned derivation; return to the source if such derivations are explicitly assessed.

In the examination. For a recognition answer name the condition and consequence; deriving Maxwell equations is unnecessary for these exercises.

Pr 48. Is a quarter-cycle phase delay alone enough to guarantee circular polarization?

Pr 49. For light from air into glass of index 1.5, estimate Brewster angle.

Pr 50. For unpolarized light at Brewster incidence in ordinary transparent nonmagnetic media:

(a) All incident light must be reflected

(b) Total internal reflection must occur

(c) The reflected light is linearly polarized

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Identify the complete cycle count and total time. Which ratio counts cycles per second? (2) Divide 50 cycles by 10 s for frequency. (3) Use total time divided by cycle count for period, or take the reciprocal of frequency. Check hertz and seconds respectively.

Pr 7 (p. 4) (1) Use the original frequency and wavelength to establish speed first. (2) Retain that speed because the medium remains nondispersive and unchanged. (3) Divide the established speed by 12 Hz for the new wavelength; do not reuse the old wavelength with the new frequency.

Pr 9 (p. 5) (1) The separation is one quarter of a full spatial cycle. (2) Therefore $\Delta\phi = 2\pi \times (1/4) = \pi/2$.

Pr 12 (p. 6) (1) The new distance is three times the original. Ask how a sphere area changes when radius triples. (2) Square the distance factor for the inverse-square intensity change. (3) Apply that factor to 18 $\mathrm{W/m^2}$ and retain intensity units; this is not a total-power calculation.

Pr 16 (p. 8) (1) Draw the fixed endpoints and the single-loop fundamental. (2) The 1 m string contains half a wavelength. Find the full wavelength before using the 40 m/s speed. (3) Divide speed by wavelength for the fundamental frequency, then use the harmonic relationship for the second harmonic.

Pr 20 (p. 9) (1) The supplied frequency ratio allows comparison without calculating Planck constant. (2) Photon energy follows the same ratio as frequency. (3) Vacuum wavelength follows the reciprocal frequency ratio. State which light has the larger energy and which has the shorter wavelength.

Pr 24 (p. 11) (1) Confirm that light travels from glass index 1.5 into air index 1.0. (2) Divide the lower index by the higher index to obtain the critical-angle sine. (3) Take inverse sine and report an angle from the normal. Check calculator degree mode if using degrees.

Pr 28 (p. 12) (1) Assign positive focal length to the converging lens and positive distance to the real object. (2) Subtract the reciprocal of 24 cm from the reciprocal of 12 cm, then invert for image distance. (3) Calculate signed magnification and interpret distance sign, orientation and size separately.

Pr 32 (p. 14) (1) Identify the objective focal length and eyepiece focal length. (2) Divide 100 cm by 5 cm for initial magnification magnitude. (3) Replace only the denominator with 10 cm and compare ratios. Keep image orientation separate from the requested magnitude.

Pr 36 (p. 15) (1) Distinguish slit separation from slit width; this is a double-slit spacing question. (2) Convert 500 nm and 0.50 mm to metres before combining them with 2 m. (3) Multiply wavelength by screen distance and divide by slit separation. Convert the final spacing to millimetres if helpful.

Pr 38 (p. 16) (1) Round-trip path change: $2 \times 250 = 500\,\mathrm{nm}$. (2) This equals one wavelength.

Pr 41 (p. 17) (1) Start with the unpolarized 80 $\mathrm{W/m^2}$ and apply the first-polarizer rule. (2) Use that transmitted intensity as the analyzer input. (3) Multiply by the squared cosine of 45 degrees. Keep the two stages visible rather than applying an extra half factor.

Pr 45 (p. 19) (1) Retain the negative focal-length sign because the lens diverges. (2) Convert -50 cm into metres. (3) Take the signed reciprocal and report dioptres. A positive result would conflict with the stated lens type.

Pr 49 (p. 20) (1) Choose the Brewster tangent relation, not the critical-angle sine relation. (2) Place glass index 1.5 over air index approximately 1.0. (3) Take inverse tangent of that ratio and measure the result from the normal. Check degree mode if reporting degrees.

Answers

Pr 2 (p. 3) No. Displacement and restoring acceleration are zero; speed is maximum.

Pr 3 (p. 3) Frequency = 50/10 = 5 Hz. Period = 10/50 = 0.20 s. Two marks each.

Pr 4 (p. 3) (a) Correct. Twice the time per cycle gives half as many cycles each second.

Pr 6 (p. 4) No. It is angular frequency; divide it by 2π to get 10 Hz.

Pr 7 (p. 4) Speed = $8 \times 0.75 = 6$ m/s. New wavelength = 6/12 = 0.50 m. Two marks each.

Pr 8 (p. 4) (b) Correct. At fixed frequency, lower speed means less distance travelled per cycle.

Pr 9 (p. 5) $\pi/2$ radians, or 90°. They are neither in phase nor exactly opposite in phase.

Pr 11 (p. 6) 2 W/m^2: divide by $3^2 = 9$.

Pr 12 (p. 6) Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.

Pr 13 (p. 6) (c) Correct. The unchanged source power is shared over each complete sphere.

Pr 15 (p. 8) It is twice the fundamental frequency in the ideal uniform-string model.

Pr 16 (p. 8) Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.

Pr 17 (p. 8) (a) Correct. One loop between neighbouring nodes occupies half a wavelength.

Pr 19 (p. 9) A has twice the photon energy and half the vacuum wavelength.

Pr 20 (p. 9) A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship.

Pr 21 (p. 9) (b) Correct. Unchanged frequency gives unchanged photon energy.

Pr 23 (p. 11) No. Total internal reflection requires travel from higher to lower refractive index.

Pr 24 (p. 11) Sine of critical angle = $\frac{1}{1}.5 = 0.6667$, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Pr 25 (p. 11) (c) Correct. The transmitted angle from the normal is 90 degrees.

Pr 27 (p. 12) It is upright and twice the object height; the sign convention and lens setup determine its image position.

Pr 28 (p. 12) 1/di = 1/12 − 1/24 = 1/24, so di = 24 cm. m = −1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.

Pr 29 (p. 12) (a) Correct. A screen intercepts the rays where they form the image.

Pr 31 (p. 14) It halves the magnification magnitude.

Pr 32 (p. 14) Initial magnitude = 100/5 = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.

Pr 33 (p. 14) (b) Correct. The eyepiece focal length is the denominator of the ratio.

Pr 35 (p. 15) It halves.

Pr 36 (p. 15) Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = $(0.0000005 \times 2)/0.0005 = 0.002$ m = 2 mm. Credit both conversions, calculation and result.

Pr 37 (p. 15) (c) Correct. Wavelength and slit separation set directions; screen distance converts direction to position.

Pr 38 (p. 16) The path changes by 500 nm, so one fringe passes a fixed observation point.

Pr 40 (p. 17) Zero intensity in the ideal model, because $\cos^2 90° = 0$.

Pr 41 (p. 17) First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.

Pr 42 (p. 17) (a) Correct. The input is already aligned, so the squared cosine of zero is one.

Pr 44 (p. 19) −2 D, since −50 cm = −0.50 m.

Pr 45 (p. 19) Convert to −0.50 m. Power=$1/(-0.50)$=−2 dioptres. Credit conversion, formula, sign and unit.

Pr 46 (p. 19) (b) Correct. The smaller focal-length magnitude has the larger reciprocal magnitude.

Pr 48 (p. 20) No. The two orthogonal components must also have equal amplitudes.

Pr 49 (p. 20) Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Pr 50 (p. 20) (c) Correct. The reflected component lacks one linear polarization component.

Optics and Waves, Test A

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which wave can propagate through vacuum?

   (a) A water wave
   (b) Light
   (c) A stretched-string wave
   (d) Sound in air

2. With wave speed fixed, doubling frequency makes wavelength:

   (a) Half as large
   (b) Unchanged
   (c) Four times as large
   (d) Twice as large

3. Group velocity describes the motion of:

   (a) The wave-packet envelope
   (b) The source itself
   (c) One particle only
   (d) A stationary node

4. Two tones at 440 and 443 Hz produce a beat frequency of:

   (a) 1320 Hz
   (b) 883 Hz
   (c) 3 Hz
   (d) 441.5 Hz

5. At the same medium, which has greater photon energy?

   (a) Lower-frequency light
   (b) Both always equal
   (c) Higher-frequency light
   (d) Longer-wavelength light

6. Light enters glass from air. Which quantity remains unchanged?

   (a) Ray direction in every case
   (b) Speed
   (c) Wavelength
   (d) Frequency

Part II. Show your reasoning.

1. A source makes 50 complete oscillations in 10 s. Find frequency and period.

2. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

3. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

4. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

5. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

6. Find the critical angle for glass of index 1.5 into air of index 1.0.

Optics and Waves, Test B

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. An image with magnification -2 is:

   (a) Upright and smaller
   (b) Inverted and twice as tall
   (c) Virtual in every case
   (d) Upright and twice as tall

2. In a simple astronomical telescope, the objective usually has:

   (a) A longer focal length than the eyepiece
   (b) Zero focal length
   (c) The same role as a polarizer
   (d) No effect on the image

3. If double-slit separation increases while wavelength and screen distance stay fixed, fringe spacing:

   (a) Becomes negative
   (b) Increases
   (c) Stays fixed
   (d) Decreases

4. Two ideal crossed polarizers transmit from an initially unpolarized beam:

   (a) One quarter
   (b) Zero
   (c) All its intensity
   (d) Half

5. The unit of optical lens power is:

   (a) Hertz
   (b) Watt
   (c) Dioptre
   (d) Newton

6. A quarter-wave plate acts primarily by changing the relative:

   (a) Charge of light
   (b) Speed of the source
   (c) Mass of photons
   (d) Phase of orthogonal components

Part II. Show your reasoning.

1. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

2. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

3. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

4. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

5. A diverging lens has focal length -50 cm. Find its optical power.

6. For light from air into glass of index 1.5, estimate Brewster angle.

Optics and Waves, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) Light is electromagnetic and does not require a material medium.

Written part. Frequency = 50/10 = 5 Hz. Period = 10/50 = 0.20 s. Two marks each. (Section: p. 2.)

Question 2. (a) Wavelength equals speed divided by frequency.

Written part. Speed = 8 × 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each. (Section: p. 3.)

Question 3. (a) The group is the envelope formed by combining nearby frequencies.

Written part. Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold. (Section: p. 5.)

Question 4. (c) Beat frequency is the absolute difference.

Written part. Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning. (Section: p. 7.)

Question 5. (c) Photon energy is proportional to frequency.

Written part. A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship. (Section: p. 8.)

Question 6. (d) The boundary does not change the frequency of the incident oscillation.

Written part. Sine of critical angle = $\frac{1}{1}.5 = 0.6667$, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation. (Section: p. 9.)

Optics and Waves, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) The minus sign indicates inversion and magnitude 2 gives the size ratio.

Written part. $1/\mathrm{di} = 1/12 - 1/24 = 1/24$, so di = 24 cm. m = −1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size. (Section: p. 11.)

Question 2. (a) The long-focus objective and short-focus eyepiece give a large angular-magnification ratio.

Written part. Initial magnitude = 100/5 = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification. (Section: p. 13.)

Question 3. (d) Fringe spacing is inversely proportional to slit separation.

Written part. Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = $(0.0000005 \times 2)/0.0005 = 0.002$ m = 2 mm. Credit both conversions, calculation and result. (Section: p. 14.)

Question 4. (b) Their axes differ by 90 degrees and $\cos^2 90^\circ$ is zero.

Written part. First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage. (Section: p. 16.)

Question 5. (c) Lens power is inverse focal length in metres, expressed in dioptres.

Written part. Convert to −0.50 m. Power=$1/(-0.50)$=−2 dioptres. Credit conversion, formula, sign and unit. (Section: p. 18.)

Question 6. (d) A controlled phase delay changes the polarization state.

Written part. Tangent of Brewster angle=1.5/1.0. The angle is $\arctan(1.5)$=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle. (Section: p. 19.)

Facts at a glance

1. Wave language and simple harmonic motion p. 2

$$T = 1/f$$
$$f = N/t$$
$$a = -\omega^2 x$$

2. Read a travelling wave without deriving it p. 3

$$v = f\lambda$$
$$k = 2\pi/\lambda$$
$$\omega = 2\pi f$$
$$y = A\sin(kx - \omega t + \phi)$$

3. Wavefronts, spreading and group velocity p. 5

$$I \propto A^2$$
$$I_{\text{spherical}} \propto 1/r^2$$

4. Superposition, standing waves and beats p. 7

$$f_{\text{beat}} = |f_1 - f_2|$$
$$\lambda_n = 2L/n \text{ (string fixed at both ends)}$$

5. Light, photons and the electromagnetic picture p. 8

$$E_\gamma = hf = hc/\lambda_0$$
$$\lambda_0 = \text{vacuum wavelength}$$
$$c = 3.00 \times 10^8 \text{ m/s}$$

6. Reflection, refraction and total internal reflection p. 9

$$\theta_i = \theta_r$$
$$n = c/v$$
$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$
$$\sin\theta_c = n_2/n_1$$

7. Mirrors, lenses and image signs p. 11

$$1/f = 1/d_o + 1/d_i$$
$$m = -d_i/d_o$$

8. Optical systems, prisms and instruments p. 13

$$m_{\text{total}} = m_1 m_2$$
$$|M_{\text{telescope}}| = f_o/f_e \text{ (normal adjustment)}$$

9. Interference and diffraction p. 14

$$\Delta y = \lambda L/d$$
$$d \sin\theta = m\lambda \text{ (double-slit bright)}$$
$$a \sin\theta = m\lambda \text{ (single-slit dark, } m \neq 0\text{)}$$

10. Polarization and Malus law p. 16

$$I = I_0 \cos^2\theta$$
$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

11. Scattering, apparent depth and lens power p. 18

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$
$$d_{\text{apparent}} \approx d_{\text{real}}/n$$
$$P_{\text{lens}} = 1/f \text{ (f in metres)}$$

12. Recognize advanced optics without long derivations p. 19

$$\tan\theta_B = n_2/n_1$$
$$E = cB \text{ (plane wave in vacuum)}$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Wave language and simple harmonic motion

N	number of complete cycles (dimensionless)
t	total elapsed time (s)
f	cycles per second (Hz)
T	time for one cycle (s)
ω	angular frequency (rad/s)

2. Read a travelling wave without deriving it

A	amplitude or maximum displacement (m)
k	wave number (rad/m)
ω	angular frequency (rad/s)
λ	wavelength (m)
f	frequency (Hz)

3. Wavefronts, spreading and group velocity

I	energy-transfer rate per unit area (W/m^2)
A	wave amplitude (depends on disturbance)
r	distance from point source (m)

4. Superposition, standing waves and beats

f_{beat}	number of loudness cycles per second (Hz)
L	vibrating string length (m)
n	harmonic number, 1, 2, 3, ... (dimensionless)
λ_n	wavelength of harmonic n (m)

5. Light, photons and the electromagnetic picture

E_γ	energy of one photon (J)
h	Planck constant (J s)
f	light frequency (Hz)
λ	vacuum wavelength (m)

6. Reflection, refraction and total internal reflection

n	refractive index (dimensionless)
c	vacuum light speed (m/s)
v	light speed in medium (m/s)
θ_c	critical incidence angle (degrees or radians)

7. Mirrors, lenses and image signs

f	signed focal length (cm or m consistently)
d_o	object distance (same length unit as f)
d_i	signed image distance (same length unit as f)
m	transverse magnification (dimensionless)

8. Optical systems, prisms and instruments

f_o	objective focal length (cm or m consistently)
f_e	eyepiece focal length (same length unit as fo)
M	angular magnification (dimensionless)
m_1, m_2	transverse magnifications of successive stages (dimensionless)

9. Interference and diffraction

Δy	double-slit fringe spacing (m)
L	slit-to-screen distance (m)
d	separation of double slits (m)
a	single-slit width (m)
λ	wavelength (m)

10. Polarization and Malus law

I_0	intensity incident on the analyzer (W/m^2 or stated units)
I	intensity transmitted by analyzer (same as I0)
θ	angle between incident polarization and analyzer axis (degrees or radians)

11. Scattering, apparent depth and lens power

d_{real}	actual depth (m)
d_{apparent}	apparent depth (m)
n	liquid refractive index relative to air approximation (dimensionless)
P	lens optical power ($D = m^{-1}$)
f	signed focal length (m)

12. Recognize advanced optics without long derivations

θ_B	Brewster incidence angle from the normal (degrees or radians)
n_1, n_2	incident and transmitted medium indices (dimensionless)
E	plane-wave electric-field magnitude (V/m)
B	plane-wave magnetic-field magnitude (T)

Sources

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

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d separation of double slits, 14
d_i signed image distance, 12
d_o object distance, 12
d_{apparent} apparent depth, 18
d_{real} actual depth, 18
E plane-wave electric-field magnitude, 20
E_γ energy of one photon, 8
f cycles per second, 2
f frequency, 4
f light frequency, 8
f signed focal length, 12, 18
f_e eyepiece focal length, 13
f_o objective focal length, 13
f_{beat} number of loudness cycles per second, 7
h Planck constant, 8
I energy-transfer rate per unit area, 5
I intensity transmitted by analyzer, 16
I_0 intensity incident on the analyzer, 16
k wave number, 4
L slit-to-screen distance, 14
L vibrating string length, 7
m transverse magnification, 12
m_1, m_2 transverse magnifications of successive stages, 13
M angular magnification, 13
n harmonic number, 1, 2, 3, ..., 7
n liquid refractive index relative to air approximation, 18
n refractive index, 10
n_1, n_2 incident and transmitted medium indices, 20
N number of complete cycles, 2
P lens optical power, 18
r distance from point source, 5
t total elapsed time, 2
T time for one cycle, 2
v light speed in medium, 10
θ angle between incident polarization and analyzer axis, 16
θ_B Brewster incidence angle from the normal, 20
θ_c critical incidence angle, 10
λ vacuum wavelength, 8
λ wavelength, 4, 14
λ_n wavelength of harmonic n, 7
ω angular frequency, 2, 4
Δy double-slit fringe spacing, 14